

ATAKPAME, Togo

WMO: 653760

Lat: 7.583N

Lon: 1.117E

Elev: 402

StdP: 96.59

Time Zone: 0.00 (E00)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.6	21.0	0.2	4.0	26.9	2.5	4.7	27.5	4.3	22.8	3.9	23.3	1.8	280	0.259

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.1	34.9	22.7	34.1	22.8	33.5	22.9	26.0	31.5	25.6	30.9	25.2	30.4	1.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.5	20.5	28.7	24.1	20.0	28.2	23.8	19.6	27.8	82.8	31.5	81.3	31.3	79.8	30.6	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 4.2	3.5	3.3	DB	19.1	36.6	1.8	0.7	17.8	37.0	16.8	37.4	15.8	37.8	14.5	38.3	(4)
(5)			WB	10.3	27.7	1.5	1.1	9.2	28.5	8.4	29.1	7.5	29.7	6.4	30.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	26.7	27.7	28.7	28.4	27.6	26.9	25.8	24.8	24.5	25.1	26.0	27.3	27.6	(6)
(7)		DBStd	1.70	1.19	0.97	1.13	1.24	1.27	1.11	0.87	0.84	0.87	0.94	0.81	0.90	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6087	548	524	571	527	522	473	460	449	452	496	518	545	(10)
(11)		CDD18.3	3045	290	291	313	277	264	223	202	191	202	237	268	287	(11)
(12)		CDH23.3	27822	3219	3485	3497	2861	2468	1654	1112	996	1204	1796	2540	2989	(12)
(13)	CDH26.7	10635	1406	1726	1642	1173	897	450	170	122	224	531	1039	1257	(13)	
(14)	Wind	WSAvg	1.7	1.8	1.8	1.9	1.9	1.6	1.6	1.7	1.5	1.5	1.6	1.6	(14)	
(15)	Precipitation	PrecAvg	1356	8	28	77	128	148	196	218	205	202	124	20	10	(15)
(16)		PrecMax	2067	60	120	192	284	324	339	435	366	380	290	98	114	(16)
(17)		PrecMin	767	0	0	0	23	53	68	52	44	11	15	0	0	(17)
(18)		PrecStd	283	13	30	52	56	57	69	96	87	90	65	23	21	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	36.0	36.0	34.2	33.3	31.8	30.1	29.7	30.5	32.0	33.6	34.1	(19)
(20)			MCWB	21.2	22.3	23.3	24.9	25.3	25.2	24.3	24.1	24.4	24.8	23.7	22.0	(20)
(21)		2%	DB	33.9	35.0	34.7	33.3	32.2	30.7	29.2	28.7	29.5	31.1	32.8	33.2	(21)
(22)			MCWB	21.5	22.2	23.6	25.0	25.1	24.7	24.2	24.0	24.5	24.7	23.6	21.8	(22)
(23)		5%	DB	33.0	34.2	33.9	32.5	31.5	30.0	28.5	28.0	28.8	30.3	32.1	32.4	(23)
(24)			MCWB	21.4	22.1	23.6	25.0	25.0	24.6	24.1	23.8	24.3	24.6	23.5	21.4	(24)
(25)		10%	DB	32.1	33.5	33.1	31.7	30.7	29.2	27.7	27.3	28.1	29.5	31.4	31.7	(25)
(26)			MCWB	20.9	22.0	23.8	24.9	24.8	24.4	23.8	23.6	24.1	24.4	23.3	21.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.7	25.5	26.2	26.7	26.4	25.7	25.2	25.0	25.5	25.7	25.5	25.1	(27)
(28)			MADB	31.5	32.7	32.9	31.7	31.2	30.3	28.8	28.2	28.7	30.4	31.3	31.6	(28)
(29)		2%	WB	24.1	24.9	25.5	26.0	25.8	25.2	24.7	24.5	24.9	25.2	25.0	24.4	(29)
(30)			MADB	30.5	31.7	32.0	31.6	30.9	29.6	28.2	27.6	28.6	29.9	30.8	30.5	(30)
(31)		5%	WB	23.6	24.5	25.0	25.6	25.4	24.9	24.4	24.1	24.5	24.9	24.6	23.8	(31)
(32)			MADB	29.4	30.9	31.5	31.0	30.4	29.1	27.7	27.1	28.1	29.4	30.3	29.4	(32)
(33)		10%	WB	23.1	24.1	24.6	25.1	25.0	24.5	24.0	23.8	24.2	24.6	24.2	23.3	(33)
(34)			MADB	28.5	30.0	30.8	30.5	29.9	28.5	27.2	26.6	27.5	28.8	29.7	28.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.0	10.1	9.9	8.6	7.9	7.1	6.1	6.0	6.5	7.8	9.4	9.0	(35)
(36)			MCDBR	10.5	11.2	10.9	9.6	9.0	8.1	7.3	7.1	7.5	8.9	10.0	10.2	(36)
(37)			MCWBR	3.6	3.3	3.1	3.4	3.2	3.2	3.0	3.1	3.2	3.6	2.8	3.4	(37)
(38)		5% WB	MCDBR	10.3	10.3	10.0	8.9	8.5	7.7	6.6	6.4	7.0	8.3	9.4	9.7	(38)
(39)			MCWBR	3.3	3.2	3.5	3.7	3.5	3.4	3.1	3.1	3.3	3.6	3.2	3.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.755	0.778	0.806	0.647	0.545	0.501	0.479	0.473	0.468	0.518	0.574	0.643	(40)	
(41)		taud	1.562	1.546	1.497	1.800	2.016	2.163	2.251	2.262	2.282	2.100	1.903	1.754	(41)	
(42)		Ebn at Noon	616	621	612	708	766	791	813	832	848	805	747	686	(42)	
(43)		Edh at Noon	278	290	307	224	176	149	138	139	138	164	196	226	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.11	5.17	5.41	5.48	5.25	4.72	4.22	3.95	4.41	5.16	5.29	5.00	(44)	
(45)		RadStd	0.13	0.15	0.21	0.18	0.16	0.21	0.17	0.20	0.17	0.15	0.11	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+0.28	+0.31	N/A	+0.36	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page